الإمارات العربية المتحدة
وزارة التربية والتعليم

الميثاق الوطني للتعليم

WEEK 6 · LESSONS 2-4 · 2-5 · 3-1

Weekly Quiz Preparation

Grade 12 Advanced · Mathematics

كتيّب التجهّز للاختبار الأسبوعي القصير

1 2 3 4 5

5	10	40	16
objectives أهداف	marks علامة	minutes دقيقة	practice questions تدريبًا

Mr. Yousef Al Yousef · **Friday 9 October 2026**

One page per objective · every solution written out step by step

Quiz map

5 objectives = 5 recipes · 5 أهداف = 5 وصفات

1 **Zero of a function** نظرية القيمة المتوسطة
Continuous + opposite signs → a zero · bisect to narrow it

2 **Limit at infinity / infinite limit** نهايات اللانهاية
Divide by the highest power · read the sign near a

3 **Asymptotes** خطوط التقارب
Vertical: denominator 0 · horizontal: x → ±∞ · slant: long division

4 **Secant to tangent** من القاطع إلى المماس
m = $\frac{f(a+h)-f(a)}{h}$, then h → 0 · different one-sided slopes: no tangent

5 **Equation of the tangent line** معادلة المماس
f(a) → difference quotient → m → y − f(a) = m(x − a)

1 · Learn
Read the idea and the recipe
اقرأ الفكرة والوصفة

2 · Watch
Cover the solution, try it, then compare
غطِّ الحل وحاول ثم قارن

3 · Try
Do “Your turn”, check at the back
حلّ التدريبات وصحّح

1 The Intermediate Value Theorem

نظرية القيمة المتوسطة

Lesson 2-4 · Part 3 · pp. 93–94
MAT.6.01.04.001

This is how you show that a function has a zero without solving f(x) = 0.

If f is continuous on [a, b] and f(a), f(b) have opposite signs, then f has a zero in (a, b).

إذا كانت f متصلة على [a, b] وإشارتا f(a) وf(b) مختلفتان فلها صفر في (a, b).

Recipe الوصفة

1. **Say: f is continuous on [a, b] (a polynomial)**
   f متصلة (كثيرة حدود)
2. **Compute f(a) and f(b)**
   احسب f(a) وf(b)
3. **Opposite signs → a zero in (a, b)**
   إشارتان مختلفتان ← صفر
4. **Bisection: test the midpoint, keep the half with the sign change**
   التنصيف: جرّب المنتصف

Watch out

- Always state that f is continuous, then the two signs.
- Same signs guarantee nothing: no zero is promised.

Tool. Corollary 4.2: f(a) · f(b) < 0 and f continuous ⇒ a zero in (a, b)

Worked example Book exercise 33(a)

Use the Intermediate Value Theorem to verify that $f(x) = x^2 - 7$ has a zero in $[2, 3]$. Then use the method of bisections twice.

1. f is a polynomial → continuous on [2, 3]; compute the ends

   $f(2) = 2^2 - 7 = -3 < 0, \qquad f(3) = 3^2 - 7 = 2 > 0$

2. opposite signs → a zero in (2, 3) (Corollary 4.2)

   $f(2)\,f(3) = -6 < 0$

3. bisect: the midpoint is 2.5

   $f(2.5) = 6.25 - 7 = -0.75 < 0 \;\Rightarrow\; [2.5,\ 3]$

4. bisect again: the midpoint is 2.75

   $f(2.75) = 7.5625 - 7 = 0.5625 > 0 \;\Rightarrow\; [2.5,\ 2.75]$

Your turn · solve in your notebook · solutions at the back

1. Book exercise 34(a) Verify with the Intermediate Value Theorem that $f(x) = x^3 - 4x - 2$ has a zero in $[2, 3]$. Then bisect until the interval has length $\frac{1}{4}$.
2. Book exercise 35 Verify that $f(x) = \cos x - x$ has a zero in $[0, 1]$ (radians), and take one step of bisection.
3. Book exercise 36 Verify that $f(x) = e^x + x$ has a zero in $[-1, 0]$, and take one step of bisection.

2 Limits involving infinity

نهايات تتضمن اللانهاية

Lesson 2-5 · Part 1 · pp. 98–106
MAT.6.01.03.001

Two jobs: a limit that blows up (x → a), and a limit as x → ∞.

At infinity, divide top and bottom by the highest power of x: every $1/x^n$ tends to 0.

عند اللانهاية اقسم البسط والمقام على أعلى قوة لـ x: كل $1/x^n \leftarrow 0$.

Recipe الوصفة

1. **Substitute: denominator → 0, numerator not → an infinite limit**
   المقام ← 0: نهاية لا نهائية
2. **Infinite limit: read the sign on each side**
   اقرأ الإشارة من كل جهة
3. **x → ±∞: divide by the highest power of x**
   اقسم على أعلى قوة
4. **A function of an expression: find the inside limit first**
   ابدأ بنهاية ما بالداخل

Watch out

- "∞ / ∞" is not an answer: divide first.
- An infinite limit needs its sign: +∞ or −∞.

Tool. $1/x^n \to 0$ as $x \to \pm\infty$ · $\tan^{-1} x \to \pm\pi/2$ · e^u → 0 as u → −∞

Worked example Book exercise 9

Determine $\lim_{x\to\infty} \dfrac{x^2+3x-2}{3x^2+4x-1}$.

1. substitution gives ∞/∞: not enough

$$\frac{x^2+3x-2}{3x^2+4x-1} \to \frac{\infty}{\infty} \text{ (no answer yet)}$$

2. divide top and bottom by x^2

$$= \lim_{x\to\infty} \frac{1+\frac{3}{x}-\frac{2}{x^2}}{3+\frac{4}{x}-\frac{1}{x^2}}$$

3. $1/x$ and $1/x^2 \to 0$

$$= \frac{1+0-0}{3+0-0} = \frac{1}{3}$$

Your turn · solve in your notebook · solutions at the back

1. Book exercise 10 Determine $\lim_{x\to\infty} \dfrac{2x^2-x+1}{4x^2-3x-1}$.
2. Book exercise 12 Determine $\lim_{x\to\infty} \dfrac{2x^2-1}{4x^3-5x-1}$.
3. Book exercise 20 Determine $\lim_{x\to\infty} \sin\left(\tan^{-1} x\right)$.
4. Ministry style Determine $\lim_{x\to 4^-} \dfrac{3}{x-4}$.

3 Vertical, horizontal and slant asymptotes
خطوط التقارب الرأسية والأفقية والمائلة

Lesson 2-5 · Part 2 · pp. 98–106
MAT.6.01.03.002

Three kinds of line a graph can approach: x = a, y = b, and y = mx + c.

Vertical: the denominator is 0. Horizontal: the limit as x → ±∞. Slant: divide, the quotient is the line.

رأسي: المقام = 0. أفقي: النهاية عند ±∞. مائل: اقسم والخارج هو المستقيم.

Recipe الوصفة

1. **Cancel common factors first**
   اختصر العوامل المشتركة
2. **Vertical: zeros of the denominator, then the sign on each side**
   أصفار المقام ثم الإشارة
3. **Horizontal: lim as x → ±∞**
   النهاية عند ±∞
4. **Slant (degree top = degree bottom + 1): long division**
   القسمة المطولة

Watch out

- The numerator must not be 0 at the vertical asymptote.
- Slant needs the top one degree higher: divide, do not guess.

Tool. deg top < deg bottom ⇒ y = 0 · equal degrees ⇒ y = ratio of leading coefficients

Worked example Book exercise 23(b)

Determine all horizontal and vertical asymptotes of $f(x) = \dfrac{x^2}{4 - x^2}$.

1. vertical: the denominator is 0

$$4 - x^2 = 0 \Rightarrow x = -2,\ x = 2$$

2. the signs near x = 2 (numerator → 4)

$$x \to 2^-:\ \frac{4}{0^+} \to +\infty, \qquad x \to 2^+:\ \frac{4}{0^-} \to -\infty$$

3. the signs near x = −2

$$x \to -2^+:\ \frac{4}{0^+} \to +\infty, \qquad x \to -2^-:\ \frac{4}{0^-} \to -\infty$$

4. horizontal: divide by x^2, $x \to \pm\infty \to y = -1$

$$\lim_{x \to \pm\infty} \frac{x^2}{4 - x^2} = \lim \frac{1}{\frac{4}{x^2} - 1} = -1$$

Your turn · solve in your notebook · solutions at the back

1. Book exercise 29 Determine all vertical and slant asymptotes of $y = \dfrac{x^3}{4 - x^2}$.
2. Book exercise 30 Determine all vertical and slant asymptotes of $y = \dfrac{x^2 + 1}{x - 2}$.
3. Book exercise 23(a) Determine all horizontal and vertical asymptotes of $f(x) = \dfrac{x}{4 - x^2}$.

4 From secant to tangent

من القاطع إلى المماس

The tangent is the limit of the secant lines: the slope is a limit of difference quotients.

The tangent slope is the limit of the secant slopes: m = lim (h → 0) [f(a + h) − f(a)] / h.

ميل المماس = نهاية ميول القواطع عندما h ← 0.

Recipe الوصفة

1. **Secant slope between x = a and x = a + h**
   ميل القاطع
2. **Expand and simplify so that h cancels**
   بسّط لينحذف h
3. **Let h → 0: that is the tangent slope**
   h ← 0: ميل المماس
4. **If the one-sided slopes differ: no tangent line**
   إن اختلفت الميول: لا مماس

Watch out

- Divide by h only after you have simplified.
- Two different one-sided slopes mean no tangent line.

Worked example Method of Exercises 9–12

For $f(x) = x^2 + 5x$, find the slope of the secant line between $x = 1$ and $x = 1 + h$, then the slope of the tangent line at $x = 1$.

1. the point: f(1)

$$f(1) = 1 + 5 = 6$$

2. f(1 + h): expand

$$f(1 + h) = (1 + h)^2 + 5(1 + h) = 6 + 7h + h^2$$

3. secant slope = [f(1 + h) − f(1)] / h

$$m_{sec} = \frac{(6 + 7h + h^2) - 6}{h} = \frac{7h + h^2}{h} = 7 + h$$

4. let h → 0: the tangent slope

$$m_{tan} = \lim_{h \to 0}(7 + h) = 7$$

Your turn · solve in your notebook · solutions at the back

1. Method of Exercises 9–12 For $f(x) = x^2 - x$, find the slope of the secant line between $x = 2$ and $x = 2 + h$, then the slope of the tangent line at $x = 2$.
2. Book exercise 23 Explain why the graph of $f(x) = |x - 1|$ has no tangent line at $x = 1$.
3. Method of Exercises 9–12 For $f(x) = 2x^2 + 1$, find the slope of the tangent line at $x = -1$ using the secant slopes.

5 The equation of the tangent line
معادلة المماس

Definition 1.1 gives the slope; the point-slope form gives the line.

The tangent line at x = a is y − f(a) = m(x − a), where m = lim (h → 0) [f(a + h) − f(a)] / h.

معادلة المماس: y − f(a) = m(x − a) حيث m نهاية خارج القسمة.

Recipe الوصفة

1. **Find f(a): the point (a, f(a))** — أوجد النقطة
2. **Difference quotient: expand, rationalize or use a common denominator** — خارج القسمة وبسّطه
3. **m = the limit as h → 0** — m = النهاية
4. **Write y − f(a) = m(x − a), then simplify** — اكتب المعادلة

Watch out

- Show the limit, not only the slope.
- Finish with the equation of the line, not just m.

Tool. Square root in the quotient: multiply by the conjugate · a fraction: common denominator first

Worked example Book exercise 1

Use Definition 1.1 to find an equation of the tangent line to $y = x^2 - 2$ at $x = 1$.

1. the point: f(1)

$$f(1) = 1 - 2 = -1 \Rightarrow (1, -1)$$

2. the difference quotient

$$\frac{f(1+h) - f(1)}{h} = \frac{(1+h)^2 - 2 + 1}{h} = \frac{2h + h^2}{h} = 2 + h$$

3. the slope is the limit

$$m = \lim_{h \to 0} (2 + h) = 2$$

4. point-slope form

$$y - (-1) = 2(x - 1) \Rightarrow y = 2x - 3$$

Your turn · solve in your notebook · solutions at the back

1. Book exercise 4 Use Definition 1.1 to find an equation of the tangent line to $y = x^3 + x$ at $x = 1$.
2. Book exercise 5 Use Definition 1.1 to find an equation of the tangent line to $y = \frac{2}{x+1}$ at $x = 1$.
3. Book exercise 8 Use Definition 1.1 to find an equation of the tangent line to $y = \sqrt{x+3}$ at $x = 1$.

Objective 1 · The Intermediate Value Theorem نظرية القيمة المتوسطة

1.1 Book exercise 34(a)

Verify with the Intermediate Value Theorem that $f(x) = x^3 - 4x - 2$ has a zero in $[2, 3]$. Then bisect until the interval has length $\frac{1}{4}$.

1. f is a polynomial → continuous; opposite signs → a zero

$$f(2) = 8 - 8 - 2 = -2 < 0, \qquad f(3) = 27 - 12 - 2 = 13 > 0$$

2. midpoint 2.5: keep the half with the sign change

$$f(2.5) = 15.625 - 10 - 2 = 3.625 > 0 \;\Rightarrow\; [2,\, 2.5]$$

3. midpoint 2.25: the interval has length 1/4

$$f(2.25) = 11.390625 - 9 - 2 = 0.390625 > 0 \;\Rightarrow\; [2,\, 2.25]$$

1.2 Book exercise 35

Verify that $f(x) = \cos x - x$ has a zero in $[0, 1]$ (radians), and take one step of bisection.

1. cos x − x is continuous; opposite signs → a zero

$$f(0) = 1 > 0, \qquad f(1) = \cos 1 - 1 \approx -0.4597 < 0$$

2. midpoint 0.5: keep [0.5, 1]

$$f(0.5) = \cos 0.5 - 0.5 \approx 0.3776 > 0 \;\Rightarrow\; [0.5,\, 1]$$

1.3 Book exercise 36

Verify that $f(x) = e^x + x$ has a zero in $[-1, 0]$, and take one step of bisection.

1. e^x + x is continuous; opposite signs → a zero

$$f(-1) = e^{-1} - 1 \approx -0.6321 < 0, \qquad f(0) = 1 > 0$$

2. midpoint −0.5: keep [−1, −0.5]

$$f(-0.5) = e^{-0.5} - 0.5 \approx 0.1065 > 0 \;\Rightarrow\; [-1,\, -0.5]$$

Objective 2 · Limits involving infinity نهايات تتضمن اللانهاية

2.1 Book exercise 10

Determine $\lim_{x\to\infty} \dfrac{2x^2 - x + 1}{4x^2 - 3x - 1}$.

1. divide by x², then every $1/x^n \to 0$

$$\lim_{x\to\infty} \frac{2 - \frac{1}{x} + \frac{1}{x^2}}{4 - \frac{3}{x} - \frac{1}{x^2}} = \frac{2}{4} = \frac{1}{2}$$

2.2 Book exercise 12

Determine $\lim_{x\to\infty} \dfrac{2x^2-1}{4x^3-5x-1}$.

1. highest power is x^3 (bottom): divide by x^3

$$\lim_{x\to\infty} \frac{\frac{2}{x}-\frac{1}{x^3}}{4-\frac{5}{x^2}-\frac{1}{x^3}} = \frac{0}{4} = 0$$

2.3 Book exercise 20

Determine $\lim_{x\to\infty} \sin\left(\tan^{-1} x\right)$.

1. the inside first: $\tan^{-1} x$ has the asymptote $y = \pi/2$

$$\tan^{-1} x \to \frac{\pi}{2} \quad (x \to \infty)$$

2. sine is continuous: put the limit inside

$$\sin\left(\tfrac{\pi}{2}\right) = 1$$

2.4 Ministry style

Determine $\lim_{x\to 4^-} \dfrac{3}{x-4}$.

1. the denominator is a small negative number

$$x \to 4^- : \; x-4 \to 0^- \text{ (small, negative)}$$

2. positive ÷ small negative → $-\infty$

$$\frac{3}{0^-} = -\infty$$

Objective 3 · Vertical, horizontal and slant asymptotes — خطوط التقارب الرأسية والأفقية والمائلة

3.1 Book exercise 29

Determine all vertical and slant asymptotes of $y = \dfrac{x^3}{4-x^2}$.

1. vertical: the denominator is 0 (numerator ≠ 0)

$$4-x^2 = 0 \;\Rightarrow\; x = -2,\; x = 2$$

2. long division: $x^3 = -x(4-x^2) + 4x$

$$\frac{x^3}{4-x^2} = -x + \frac{4x}{4-x^2}$$

3. the remainder fraction → 0: slant asymptote $y = -x$

$$\frac{4x}{4-x^2} \to 0 \;\Rightarrow\; y = -x$$

3.2 Book exercise 30

Determine all vertical and slant asymptotes of $y = \dfrac{x^2+1}{x-2}$.

1. vertical: the denominator is 0

$$x - 2 = 0 \Rightarrow x = 2 \text{ (numerator 5)}$$

2. long division

$$\frac{x^2+1}{x-2} = x + 2 + \frac{5}{x-2}$$

3. slant asymptote y = x + 2

$$\frac{5}{x-2} \to 0 \Rightarrow y = x + 2$$

3.3 Book exercise 23(a)

Determine all horizontal and vertical asymptotes of $f(x) = \dfrac{x}{4-x^2}$.

1. vertical asymptotes x = −2 and x = 2

$$4 - x^2 = 0 \Rightarrow x = \pm 2$$

2. horizontal asymptote y = 0

$$\lim_{x\to\pm\infty} \frac{x}{4-x^2} = \lim \frac{\frac{1}{x}}{\frac{4}{x^2} - 1} = 0$$

Objective 4 · From secant to tangent

من القاطع إلى المماس

4.1 Method of Exercises 9–12

For $f(x) = x^2 - x$, find the slope of the secant line between $x = 2$ and $x = 2 + h$, then the slope of the tangent line at $x = 2$.

1. the point and f(2 + h)

$$f(2) = 2, \quad f(2+h) = (2+h)^2 - (2+h) = 2 + 3h + h^2$$

2. secant slope

$$m_{sec} = \frac{3h + h^2}{h} = 3 + h$$

3. the tangent slope is 3

$$m_{tan} = \lim_{h\to 0}(3 + h) = 3$$

4.2 Book exercise 23

Explain why the graph of $f(x) = |x - 1|$ has no tangent line at $x = 1$.

1. secant slopes from the right: h > 0

$$x > 1: \quad \frac{f(1+h) - f(1)}{h} = \frac{|h|}{h} = \frac{h}{h} = 1$$

2. secant slopes from the left: h < 0

$$x < 1: \quad \frac{|h|}{h} = \frac{-h}{h} = -1$$

3. the one-sided slopes differ: no tangent line

1 and -1 differ

4.3 Method of Exercises 9–12

For $f(x) = 2x^2 + 1$, find the slope of the tangent line at $x = -1$ using the secant slopes.

1. the point and f(−1 + h)

$$f(-1) = 3, \quad f(-1+h) = 2(1 - 2h + h^2) + 1 = 3 - 4h + 2h^2$$

2. secant slope

$$m_{sec} = \frac{-4h + 2h^2}{h} = -4 + 2h$$

3. the tangent slope is −4

$$m_{tan} = \lim_{h \to 0}(-4 + 2h) = -4$$

Objective 5 · The equation of the tangent line معادلة المماس

5.1 Book exercise 4

Use Definition 1.1 to find an equation of the tangent line to $y = x^3 + x$ at $x = 1$.

1. the point

$$f(1) = 2 \Rightarrow (1, 2)$$

2. expand (1 + h)³ and simplify

$$\frac{(1+h)^3 + (1+h) - 2}{h} = \frac{4h + 3h^2 + h^3}{h} = 4 + 3h + h^2$$

3. the slope is 4: point-slope form

$$m = 4, \quad y - 2 = 4(x - 1) \Rightarrow y = 4x - 2$$

5.2 Book exercise 5

Use Definition 1.1 to find an equation of the tangent line to $y = \dfrac{2}{x+1}$ at $x = 1$.

1. the point

$$f(1) = 1 \Rightarrow (1, 1)$$

2. common denominator, then cancel h

$$\frac{\frac{2}{2+h} - 1}{h} = \frac{2 - (2+h)}{h(2+h)} = \frac{-1}{2+h}$$

3. the slope is -1/2: point-slope form

$$m = -\frac{1}{2}, \quad y - 1 = -\frac{1}{2}(x-1) \Rightarrow y = -\frac{1}{2}x + \frac{3}{2}$$

5.3 Book exercise 8

Use Definition 1.1 to find an equation of the tangent line to $y = \sqrt{x+3}$ at $x = 1$.

1. the point

$$f(1) = 2 \Rightarrow (1, 2)$$

2. rationalize the numerator

$$\frac{\sqrt{4+h} - 2}{h} = \frac{1}{\sqrt{4+h} + 2}$$

3. the slope is 1/4: point-slope form

$$m = \frac{1}{4}, \quad y - 2 = \frac{1}{4}(x-1) \Rightarrow y = \frac{1}{4}x + \frac{7}{4}$$

Ready?

Tick each line · ضع علامة

- [ ] **I can verify a zero with the Intermediate Value Theorem and bisect**
  أستطيع استخدام نظرية القيمة المتوسطة والتنصيف
- [ ] **I can find an infinite limit and a limit at infinity**
  أستطيع إيجاد نهايات اللانهاية
- [ ] **I can find vertical, horizontal and slant asymptotes**
  أستطيع إيجاد خطوط التقارب
- [ ] **I can go from secant slopes to the tangent slope**
  أستطيع الانتقال من ميول القواطع إلى ميل المماس
- [ ] **I can write the equation of a tangent line with Definition 1.1**
  أستطيع كتابة معادلة المماس

On the day

Bring a scientific calculator (radian mode for cos x). Write the first line at once: the first step already earns a mark.

Every question

Say why: “f is continuous”, the sign on each side, “by Definition 1.1”. Keep the lim sign on every line until you substitute.

One mark for every written step

لكل خطوة تكتبها علامة